

Hong Yi Huang

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Alfréd Rényi Institute of Mathematics, Reáltanoda utca 13-15., H-1053, Budapest, Hungary

Research interests

- Group theory (permutation groups, simple groups, algebraic groups)
- Algebraic combinatorics
- Representation theory

Employment

Research Fellow, HUN-REN Alfréd Rényi Institute of Mathematics, Hungary *from May 2026*

- Academic host: Professor László Pyber

Research Scholar, Southern University of Science and Technology, China *Nov 2025 – Apr 2026*

- Academic host: Professor Cai Heng Li

Research Fellow, University of St Andrews, United Kingdom *Apr – Oct 2025*

- Academic host: Professor Colva Roney-Dougal
- With a Early Career Fellowship from the London Mathematical Society
- Financial support: London Mathematical Society & Heilbronn Institute for Mathematical Research

Education

University of Bristol (UoB), United Kingdom *Jan 2021 – Mar 2025*

- PhD in Mathematics
- Supervisor: Professor Tim Burness
- Funding: China Scholarship Council
- Thesis: *Bases for permutation groups and related problems*

Southern University of Science and Technology (SUSTech), China *Sept 2016 – July 2020*

- Bachelor of Science
- Major: Mathematics and Applied Mathematics
- Supervisor: Professor Cai Heng Li
- Thesis: *On valency problems of Saxl graphs of almost simple primitive groups with soluble stabiliser*

Distinctions

LMS Early Career Fellowship (6 fellows in 2025) *2025*

Oberwolfach Leibniz Graduate Students grant, Mathematical Research Institute Oberwolfach *2025*

Alumni Grant: Conference Travel Fund (£300), UoB *2024*

China Scholarship Council (full PhD scholarship at UoB) *2021–25*

Journal publications and preprints

- [10] H.Y. Huang and C.M. Roney-Dougall, *Greedy bases and relational complexity of diagonal type groups*, submitted (2026). arxiv:2605.16032
- [9] S.D. Freedman, H.Y. Huang, M. Lee and K. Rekvényi, *On the generalised Saxl graphs of permutation groups*, **Algebr. Comb.**, to appear. arxiv:2410.22613
- [8] T.C. Burness and H.Y. Huang, *On the intersections of nilpotent subgroups in simple groups*, **Proc. London Math. Soc.** 132 (2026), Paper No. e70134, 48 pp. doi:10.1112/plms.70134
- [7] T.C. Burness and H.Y. Huang, *On the intersections of Sylow subgroups in almost simple groups*, **J. Algebra** 690 (2026), 596–631. doi:10.1016/j.jalgebra.2025.11.003
- [6] H.Y. Huang, C.H. Li and Y.L. Xie, *Fixers and derangements of finite permutation groups*, **J. Algebraic Combin.** 61 (2025), Paper No. 57, 44 pp. doi:10.1007/s10801-025-01425-3
- [5] H.Y. Huang, C.H. Li and Y.Z. Zhu, *On finite permutation groups of rank three*, **J. Algebra** 666 (2025), 703–732. doi:10.1016/j.jalgebra.2024.11.028
- [4] H.Y. Huang, *Base sizes of primitive groups of diagonal type*, **Forum Math. Sigma** 12 (2024), Paper No. e2, 43 pp. doi:10.1017/fms.2023.121
- [3] T.C. Burness and H.Y. Huang, *On base sizes for primitive groups of product type*, **J. Pure Appl. Algebra** 227 (2023), Paper No. 107228, 43 pp. doi:10.1016/j.jpaa.2022.107228
- [2] T.C. Burness and H.Y. Huang, *On the Saxl graphs of primitive groups with soluble stabilisers*, **Algebr. Comb.** 5 (2022), 1053–1087. doi:10.5802/alco.238
- [1] J. Chen and H.Y. Huang, *On valency problems of Saxl graphs*, **J. Group Theory** 25 (2022), 543–577. doi:10.1515/jgth-2021-0123

Work in progress

- *On chiral generating pairs of finite simple groups.*
- *The explicit factorisations of alternating and symmetric groups*, with J. Chen and J. Zhu.
- *On TI-subgroups of simple groups*, with W. Di and Y.Z. Zhu.
- *On transitive subgroups of primitive groups*, with L. Wang.

Conference & seminar organisation

26th Postgraduate Group Theory Conference (PGTC), UoB

July 2025

- With Marina Anagnostopoulou-Merkouri, Pénélope Azuelos, David Guo, Merlin Haith Rowlatt and Sarah Martin
- Funded by the London Mathematical Society, the Heilbronn Institute for Mathematical Research, Jane Street Capital, and the School of Mathematics of UoB
- Website: <https://sites.google.com/view/pgtc2025>

Bristol Junior Algebra and Combinatorics Seminar (BJACS), UoB

Feb–May 2024

- Founding co-organiser
- With Marina Anagnostopoulou-Merkouri and Merlin Haith Rowlatt

- Website: <https://sites.google.com/view/bjacs>

4th International Conference on Groups, Graphs and Combinatorics, SUSTech

Nov 2023

- With a group of people led by Cai Heng Li

Seminars on Groups and Graphs, SUSTech (online)

2021–24

- Founding co-organiser
- With Zhaochen Ding (2021–24), Yilin Xie (2021–23), Junyan Chen (2024) and Fu-Gang Yin (2024)
- Website: <https://www.gtseminar.xyz/>

Conference invitations

Finite Groups, Fusion Systems and Applications, Mathematical Research Institute Oberwolfach

Mar 2025

Young Group Theorists Workshop, SwissMAP, Les Diablerets, Switzerland

Sept 2022

Invited talks & seminars

2026

Groups, permutations and bases, University of St Andrews

5 Feb

2025

Intersections of Sylow subgroups in simple groups, Yunnan University of Finance and Economics

23 Dec

Transitive subgroups of primitive groups, Groups and Combinatorics, China Three Gorge University

25 Oct

Intersections of Sylow subgroups in simple groups, Central China Normal University

24 Oct

Intersections of Sylow subgroups in simple groups, Peking University

25 Sept

Intersections of Sylow subgroups in simple groups, Beijing Jiaotong University

24 Sept

Permutation groups, transitive subgroups and bases, Pure Maths Colloquium, University of St Andrews

17 Apr

Permutation groups, symmetry breaking and probability, Great Bay University

26 Feb

Permutation groups, symmetry breaking and probability, Guangxi University

25 Feb

2024

Permutation groups, symmetry breaking and probability, Zhejiang University

24 Dec

Bases and Saxl graphs in the past four years, Discrete Mathematics Seminar, SUSTech

12 Dec

Bases and Saxl graphs for permutation groups, BJACS, UoB

18 Sept

Bases for permutation groups, University of Milano-Bicocca, Italy

6 Sept

Permutations, bases and low rank groups, Joint Algebra Seminar, RWTH Aachen

13 June

Permutations, bases and low rank groups, Advanced Seminar on Group Theory, HHU Düsseldorf

11 June

Permutations, bases and low rank groups, Seminar Groups and Representations, Kaiserslautern-Landau

6 June

Permutations, bases and low rank groups, Algebra Seminar, UoB

19 Mar

Bases for permutation groups, Algebra Seminar, University of Warwick

29 Jan

2023

Bases for permutation groups, Algebra Seminar, University of Birmingham

30 Nov

Symmetry breaking on graphs and groups, Seminars on Discrete Mathematics, Xiamen University

26 Sept

Permutation groups: bases and Saxl graphs, Capital Normal University

16 Sept

Bases for permutation groups, SUSTech

7 Sept

Bases for permutation groups, Junior London Algebra Colloquium, Imperial College London

23 Mar

2022

Bases, distinguishing partitions and probabilistic methods, The University of Melbourne (online)

18 Aug

2021

- Groups, graphs and transitivity*, Junior Algebra Colloquium, UoB (online) 21 May
The probabilistic method in group theory, Discrete Mathematics Seminar, SUSTech 22 Apr

2020

- On valency problems of Saxl graphs*, Discrete Mathematics Seminar, SUSTech 26 Nov

Other talks

2025

- Intersections of Sylow subgroups in simple groups*, Finite Groups and their Representations, Xiamen 18 Oct
Transitive subgroups of primitive groups, Mathematical Research Institute Oberwolfach, Germany 20 Mar
Regular subgroups of primitive groups, Groups, Graphs and Combinatorics, Kunming 12 Jan

2024

- The generalised Saxl graphs of finite permutation groups*, Topics in Group Theory, Padova, Italy 11 Sept
Permutation groups of rank three, Group Theory in Florence IV, Florence, Italy 2 July

2023

- Bases for permutation groups*, 16th National Conference on Algebra, Quanzhou 18 Nov
Bases for permutation groups, 4th International Conference on Groups, Graphs and Combinatorics 11 Nov
Symmetry breaking on primitive groups, Seminars on Groups and Graphs, SUSTech 4 Mar

2022

- Base-two primitive permutation groups*, Young Group Theorists Workshop, SwissMAP, Les Diablerets 5 Sept
Base-two primitive permutation groups, Groups St Andrews, Newcastle, UK 4 Aug
Base-two primitive permutation groups and their Saxl graphs, Groups and Graphs 24 July
Base-two primitive permutation groups, 23rd Postgraduate Group Theory Conference 8 July
Bases for primitive permutation groups, Group Theory Seminar, SUSTech 24 May
Regular orbits of primitive groups on power sets, Group Theory Seminar, SUSTech 19 Feb

2021

- Base-two primitive permutation groups and their Saxl graphs*, 43rd Australasian Combinatorics Conference 13 Dec
Regular suborbits of finite primitive groups, Groups, Graphs and Combinatorics 14 Nov
Base-two primitive permutation groups and their Saxl graphs, LMS Graduate Student Meeting 8 Nov
The distinguishing number of permutation groups, Group Theory Seminar, SUSTech 9 Oct
On valency problems of Saxl graphs, Workshop on Group Actions and Transitive Graphs 2 Jan

2020

- On valency problems of Saxl graphs*, SUSTech 17 Nov

Other conferences

2026

- LMS Research School on Growth and Expansion in Groups*, University of Warwick July
Modern Flavours of Finite Groups, University of Warwick June

2025

- Conference on Algebraic Graph Theory and Combinatorics*, SUSTech Nov
Groups and Combinatorics, China Three Gorge University, Yichang, China Oct

<i>Finite Groups and their Representations</i> , Xiamen University, Xiamen, China	Oct
<i>Groups, Graphs and Combinatorics</i> , Yunnan University, Kunming, China	Jan
2024	
<i>Topics in Group Theory</i> , University of Padova	Sept
<i>Group Theory in Florence IV</i> , University of Florence	July
<i>New perspectives in Computational Group Theory</i> , University of Warwick	June
2023	
<i>16th National Conference on Algebra</i> , Quanzhou, China	Nov
<i>International Conference on Combinatorics and Finite Geometry</i> , SUSTech	Sept
<i>CMI-HIMR Summer School: Probabilistic Group Theory</i> , UoB	June
<i>Fry Conference Series: New Perspectives in Pure Mathematics</i> , UoB	Mar
2022	
<i>Groups St Andrews</i> , Newcastle University	Aug
<i>Simple groups, Representations and Applications</i> , Isaac Newton Institute, University of Cambridge	July
<i>Groups and Graphs</i> , Jiangxi University of Science and Technology (online)	July
<i>23rd Postgraduate Group Theory Conference</i> , London	July
2021	
<i>43rd Australasian Combinatorics Conference</i> , The University of Melbourne (online)	Dec
<i>Groups, Graphs and Combinatorics</i> , SUSTech (online)	Nov
<i>LMS Graduate Student Meeting</i> , London Mathematical Society (online)	Nov
<i>Workshop on Group Actions and Transitive Graphs</i> , Kunming, China	Jan
Before 2020	
<i>Workshop on Combinatorics and Graph Theory</i> , SUSTech	Oct 2020
<i>International Conference on Algebraic Combinatorics</i> , Jiaozuo, China	Sept 2019
<i>The Third International Conference on Group Actions and Transitive Graphs</i> , SUSTech	Oct 2018

Teaching

Tutorial leader, School of Mathematics, UoB

- 2023–2024 Teaching Block 2: MATH10010 Introduction to Group Theory (3 tutorial groups)
- 2021–2022 Teaching Block 1: MATH10010 Introduction to Proofs (3 tutorial groups)

Teaching assistant, Department of Mathematics, SUSTech

- 2019–2020 Spring: MA109 Advanced Linear Algebra (1 out of 3 tutorial groups)
- 2019–2020 Fall: MA107 Advanced Linear Algebra I (2 out of 2 tutorial groups)
- 2018–2019 Spring: MA104b Linear Algebra II (1 out of 4 tutorial groups)

Homework marker, Department of Mathematics, SUSTech

- 2020–2021 Spring: MA321 Group Representation Theory, and MA219-16 Elementary Number Theory
- 2019–2020 Fall: MA321 Group Representation Theory, and MA219 Abstract Algebra (H)
- 2018–2019 Spring: MAT8010 Combinatorics (PG)
- 2017–2018 Fall: MA219-16 Elementary Number Theory, and MA213-16 Mathematical Analysis

Research visits

2026: University of St Andrews (1 week, Colva Roney-Dougal)

2025: Yunnan University of Finance and Economics (1 week, Cixuan Wu); Central China Normal University (3 days, Yuanyang Zhou); Xiamen University (3 days, Jiyong Chen); Peking University (1 week, Jiping Zhang); Beijing Jiaotong University (1 day, Fu-Gang Yin); Warwick (1 day, Sean Eberhard and Gareth Tracey); Great Bay University (1 day, Weicong Li); Guangxi University (1 day, Huye Chen); Yunnan University (1 week, Lei Wang)

2024: Zhejiang University (1 week, Tao Feng); SUSTech (1 week, Cai Heng Li); Milano-Bicocca (3 days, Pablo Spiga); Warwick (1 day, Gareth Tracey); RWTH Aachen (3 days, Alice Niemeyer); HHU Düsseldorf (2 days, Benjamin Klopsch); Kaiserslautern-Landau (1 week, Gunter Malle); Imperial College London (3 days, Kamilla Rekvényi)

2023: Xiamen University (1 week, Jiyong Chen); Capital Normal University (1 week, Shaofei Du); SUSTech (1 month, Cai Heng Li)

2021: SUSTech (2 months, Qing Xiang)

2020: SUSTech (6 months, Cai Heng Li)

Professional activities

- Refereeing assignments: J. Algebra, European J. Combin., J. Group Theory, J. Combin. Theory Ser. A, Proc. Roy. Soc. Edinburgh Sect. A
- Reviewer for AMS Mathematical Reviews (4 reviews) *since 2023*
- Member of the American Mathematical Society *since 2026*
- Associate Member of the London Mathematical Society *2024–25*
- Member of the Combinatorial Mathematics Society of Australasia *2022*